UNIT 2 - COMPLETE NOTES

Real and Complex Numbers

The number system from counting numbers to complex numbers: real numbers and their properties, radicals and exponents, scientific notation, and arithmetic with complex numbers.

What this unit covers

- **2.1** Real numbers - rational and irrational numbers, the number line
- **2.2** Properties of real numbers - closure, commutative, associative, identity, inverse, distributive
- **2.3** Radicals and radicands - laws of exponents and radicals
- **2.4** Scientific notation - very large and very small numbers
- **2.5** Complex numbers - the number *i*, real and imaginary parts, conjugate
- **2.6** Operations on complex numbers - addition, subtraction, multiplication, division

How to use these notes

Read one section at a time and work through every worked example with pen and paper before looking at the solution. Yellow boxes hold the rules most often tested in board papers. The last two pages give solved examples, board-style multiple-choice questions and short questions for self-testing.

Exam focus: Board papers regularly ask for (1) identifying rational and irrational numbers, (2) verifying a property of real numbers with given values, (3) simplifying radicals, (4) converting to and from scientific notation, and (5) adding, multiplying and dividing complex numbers.

Based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 2, starting at book page 50. Notes prepared by Shah Fahad.

1. The real number system

Each set below contains the one before it, and together the rational and irrational numbers make up the **real numbers**, written as **R**.

Set	Symbol	Meaning and examples
Natural numbers	**N**	Counting numbers: 1, 2, 3, 4, ...
Whole numbers	**W**	Naturals with zero: 0, 1, 2, 3, ...
Integers	**Z**	..., -3, -2, -1, 0, 1, 2, 3, ...
Rational numbers	**Q**	Any number written as p/q, p and q integers, q not 0. Examples: 3/4, -5, 0.25, 2.333...
Irrational numbers	**Q'**	Cannot be written as p/q; decimal neither ends nor repeats. Examples: sqrt(2), sqrt(3), pi
Real numbers	**R**	All rational and irrational numbers together: R = Q union Q'

Definition - Rational number: A number that can be expressed in the form p/q, where p and q are integers and q ≠ 0.
Definition - Irrational number: A real number that is not rational.

Decimals tell the two apart

- **Terminating decimals** are rational: 0.5 = 1/2, 0.125 = 1/8.
- **Recurring (repeating) decimals** are rational: 0.333... = 1/3, 0.272727... = 3/11.
- **Non-terminating, non-recurring decimals** are irrational: 1.414213... (sqrt(2)), 3.141592... (pi).

Example 1. Classify: (a) 7/9 (b) sqrt(16) (c) sqrt(5) (d) 0.1010010001...
Solution: (a) Rational - it is in p/q form. (b) Rational - sqrt(16) = 4, an integer. (c) Irrational - 5 is not a perfect square, so sqrt(5) has a non-terminating, non-recurring decimal. (d) Irrational - the pattern of zeros grows, so the decimal never repeats in a fixed block.

Remember: Every integer is rational (write it over 1), but not every rational number is an integer. A square root is rational only when the number under the root is a perfect square.

On the number line: every real number matches exactly one point, and every point matches exactly one real number - including irrationals such as sqrt(2), between 1.4 and 1.5.

2. Properties of real numbers

For any real numbers *a*, *b* and *c*, the following properties hold. Board questions usually give you numbers and ask you to name and verify the property.

Property	Addition	Multiplication
Closure	a + b is real	a x b is real
Commutative	a + b = b + a	a x b = b x a
Associative	(a + b) + c = a + (b + c)	(a x b) x c = a x (b x c)
Identity	a + 0 = a (0 is the additive identity)	a x 1 = a (1 is the multiplicative identity)
Inverse	a + (-a) = 0	a x (1/a) = 1, for a ≠ 0

Distributive property: a x (b + c) = a x b + a x c. Multiplication distributes over addition. Example: 3 x (4 + 5) = 3 x 4 + 3 x 5 = 27.

Order and trichotomy

- For any two real numbers, exactly one is true: a < b, a = b, or a > b (**trichotomy property**).
- If a < b then a + c < b + c. Adding the same number to both sides keeps the inequality.
- Multiplying both sides by a positive number keeps the inequality; multiplying by a negative number reverses it.

Example 2. Verify the distributive property for a = 2, b = 3, c = 4.
Solution: Left side: a(b + c) = 2(3 + 4) = 2 x 7 = 14. Right side: ab + ac = 2 x 3 + 2 x 4 = 6 + 8 = 14. Both sides are equal, so the property is verified.

Example 3. Name the property: (a) 5 + 9 = 9 + 5 (b) 7 x 1 = 7 (c) 6 + (-6) = 0
Solution: (a) Commutative property of addition. (b) Multiplicative identity. (c) Additive inverse.

Watch out: Subtraction and division are **not** commutative (5 - 3 ≠ 3 - 5) and **not** associative. Division by zero is undefined, so 0 has no multiplicative inverse.

3. Radicals, radicands and exponents

> **Radical and radicand:** In the expression $\sqrt[n]{x}$, the symbol is the **radical sign**, *n* is the **index** (order) of the radical, and *x* is the **radicand** - the number under the radical sign. If no index is written, the index is 2 (a square root).

A radical is really a fractional exponent. This one fact connects the whole topic:

$$\sqrt[n]{x} = x^{1/n} \quad \text{and} \quad \sqrt[n]{x^m} = x^{m/n}$$

Laws of exponents (for real a, b and rational m, n)

Law	Example
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^2 = 2^5 = 32$
$a^m / a^n = a^{m-n}$	$5^4 / 5^2 = 5^2 = 25$
$(a^m)^n = a^{mn}$	$(3^2)^3 = 3^6 = 729$
$(ab)^n = a^n b^n$	$(2 \times 3)^2 = 4 \times 9 = 36$
$a^0 = 1\ (a \neq 0)$; $a^{-n} = 1 / a^n$	$7^0 = 1$; $2^{-3} = 1/8$

Laws of radicals

- $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$ | $\sqrt[n]{a/b} = \sqrt[n]{a} / \sqrt[n]{b}$
- $(\sqrt[n]{a})^n = a$ | $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$

Example 4. Simplify: (a) sqrt(72) (b) sqrt(12) x sqrt(3) (c) $27^{2/3}$

Solution: (a) sqrt(72) = sqrt(36 x 2) = 6 sqrt(2). (b) sqrt(12 x 3) = sqrt(36) = 6. (c) $27^{2/3}$ = (cube root of 27)^2 = 3^2 = 9.

> **Simplest radical form:** A radical is fully simplified when the radicand has no perfect-square factor (for square roots), no fraction sits under the radical, and no radical sits in a denominator. With rational exponents: $\sqrt{x^5} = x^{5/2}$ and $\sqrt[4]{y^3} = y^{3/4}$.

4. Scientific notation

Scientists write very large and very small numbers in a short standard form. A number is in **scientific notation** when it is written as:

$$a \times 10^{n} \quad \text{where } 1 \le a < 10 \text{ and } n \text{ is an integer}$$

- **Large numbers** give a positive exponent: count the places the decimal point moves left.
- **Small numbers** (less than 1) give a negative exponent: count the places the decimal point moves right.

Ordinary number	Scientific notation	Why
5,930,000	5.93×10^{6}	Decimal point moved 6 places left
299,792,458 (speed of light, m/s)	2.99792458×10^{8}	Point moved 8 places left
0.00042	4.2×10^{-4}	Point moved 4 places right

Example 6. Write in scientific notation: (a) 47,500,000 (b) 0.000065
Solution: (a) 4.75×10^{7}. (b) 6.5×10^{-5}.

Example 7. Write as an ordinary number: (a) 3.16×10^{5} (b) 8.04×10^{-3}
Solution: (a) 316,000. (b) 0.00804.

Calculating with scientific notation

Multiply the decimal parts, then add the exponents. Divide the decimal parts, then subtract the exponents. If the decimal part finishes at 10 or more (or below 1), adjust it and correct the exponent.

Example 8. $(2 \times 10^{3}) \times (4 \times 10^{5})$
Solution: $= (2 \times 4) \times 10^{3+5} = 8 \times 10^{8}$.

Example 9. $(9 \times 10^{7}) / (3 \times 10^{2})$
Solution: $= (9 / 3) \times 10^{7-2} = 3 \times 10^{5}$.

Common error: 42×10^{5} is **not** scientific notation, because 42 is not between 1 and 10. Correct form: 4.2×10^{6}.

5. Complex numbers

The equation $x^2 = -1$ has no real solution, because the square of every real number is zero or positive. Mathematicians solved this by introducing a new number:

$$i = \text{sqrt}(-1) \quad \text{so that} \quad i^2 = -1$$

> **Definition - Complex number:** A number of the form **a + bi**, where a and b are real numbers and i = sqrt(-1). The set of complex numbers is written as **C**. Here **a** is the **real part** and **b** is the **imaginary part**.

- If b = 0, the number a + 0i = a is purely real. So **every real number is also a complex number**.
- If a = 0 and b ≠ 0, the number bi is purely imaginary, e.g. 5i.
- Two complex numbers are **equal** only when their real parts are equal and their imaginary parts are equal: a + bi = c + di means a = c and b = d.

Powers of i - a cycle of four

i^1	i^2	i^3	i^4	Then it repeats
i	-1	-i	1	$i^5 = i$, $i^6 = -1$, ...

For any power of i, divide the exponent by 4 and use the remainder: $i^{23} = i^3 = -i$.

The conjugate

> **Conjugate:** The conjugate of a + bi is **a - bi** - only the sign of the imaginary part changes. The conjugate of 3 + 4i is 3 - 4i. A complex number multiplied by its conjugate always gives a real number: $(a + bi)(a - bi) = a^2 + b^2$.

Example 10. For z = -5 + 2i, write (a) the real part, (b) the imaginary part, (c) the conjugate.
Solution: (a) Real part = -5. (b) Imaginary part = 2 (note: it is 2, not 2i). (c) Conjugate = -5 - 2i.

Example 11. Find x and y if x + yi = 7 - 3i.
Solution: Equal complex numbers have equal parts, so x = 7 and y = -3.

> **Negative roots:** sqrt(-9) = sqrt(9) x sqrt(-1) = 3i. In general, sqrt(-b) = i sqrt(b) for b > 0.

6. Operations on complex numbers

Treat i like an ordinary letter while you calculate, then replace every i^2 with -1 at the end.

Addition and subtraction - combine like parts

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Example 12. (4 + 3i) + (2 - 5i) = (4 + 2) + (3 - 5)i = **6 - 2i**.
(4 + 3i) - (2 - 5i) = (4 - 2) + (3 + 5)i = **2 + 8i**.

Multiplication - expand, then use i squared = -1

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Example 13. (2 + 3i)(1 + 4i)
Solution: = 2 + 8i + 3i + $12i^2$ = 2 + 11i + 12(-1) = 2 + 11i - 12 = **-10 + 11i**.

Division - multiply top and bottom by the conjugate

Division works because the conjugate clears i from the denominator: (c + di)(c - di) = $c^2 + d^2$, a real number.

Example 14. Divide (3 + 2i) by (1 - i).
Solution: Multiply top and bottom by the conjugate of the denominator, (1 + i):
(3 + 2i)(1 + i) / (1 - i)(1 + i) = (3 + 3i + 2i + $2i^2$) / (1 + 1) = (1 + 5i) / 2 = **1/2 + (5/2)i**.

Example 15. Simplify $i^{18} + i^{11}$.
Solution: 18 divided by 4 leaves remainder 2, so i^{18} = -1. 11 divided by 4 leaves remainder 3, so i^{11} = -i. Sum = **-1 - i**.

Method checklist: Add/subtract -> match real with real, imaginary with imaginary. Multiply -> expand fully, then i^2 = -1. Divide -> conjugate of the denominator, top and bottom. Always finish in the form a + bi.

7. Solved examples - mixed practice

Example 16. Is 0.12112111211112... rational or irrational?
Solution: The decimal never ends and the blocks of 1s keep growing, so no fixed block repeats. The number is **irrational**.

Example 17. Simplify sqrt(50) + sqrt(18).
Solution: sqrt(50) = 5 sqrt(2) and sqrt(18) = 3 sqrt(2). Sum = 8 sqrt(2).

Example 18. The distance between two stars is about 94,600,000,000,000 km. Write it in scientific notation.
Solution: The decimal point moves 13 places left: **9.46×10^{13} km**.

Example 19. If z = 3 - 2i, find z multiplied by its conjugate.
Solution: Conjugate of z is 3 + 2i. Product = $3^2 + 2^2 = 9 + 4 =$ **13**, a real number - as it always is.

Example 20. Find real x and y if (x + 2) + (y - 1)i = 5 + 4i.
Solution: Equating parts: x + 2 = 5, so x = 3; y - 1 = 4, so y = 5.

Example 21. Simplify $(1 + i)^2$.
Solution: $= 1 + 2i + i^2 = 1 + 2i - 1 =$ **2i**.

Common mistakes to avoid

- Writing the imaginary part of 3 - 4i as -4i. It is **-4**; the i belongs to the number, not the part.
- Forgetting that $i^2 = -1$ when multiplying - this is the single most common lost mark in this unit.
- Calling sqrt(25) irrational. sqrt(25) = 5, which is rational.
- Leaving an answer like 30×10^4 and calling it scientific notation. The first factor must be from 1 up to (not including) 10.
- Dividing complex numbers without using the conjugate, leaving i in the denominator.

8. Test yourself - MCQs and short questions

Multiple-choice questions

1. Which of the following is an irrational number? (a) 22/7 (b) 3.1416 (c) pi (d) 0.101001000... with a repeating block of fixed length
   Answer: (c) - pi cannot be written as p/q. 22/7 is only an approximation of pi.
2. The additive inverse of -7 is: (a) 1/7 (b) 7 (c) -1/7 (d) 0
   Answer: (b)
3. sqrt(98) in simplest radical form is: (a) 2 sqrt(7) (b) 7 sqrt(2) (c) 49 sqrt(2) (d) sqrt(49 x 2) unsimplified is already simplest
   Answer: (b) - sqrt(98) = sqrt(49 x 2) = 7 sqrt(2).
4. 0.0000034 in scientific notation is: (a) 34×10^{-7} (b) 3.4×10^{-6} (c) 3.4×10^{6} (d) 0.34×10^{-5}
   Answer: (b)
5. i^{25} equals: (a) 1 (b) -1 (c) i (d) -i
   Answer: (c) - 25 divided by 4 leaves remainder 1.
6. The conjugate of -2 - 3i is: (a) 2 + 3i (b) -2 + 3i (c) 2 - 3i (d) -2 - 3i
   Answer: (b)
7. (5 + 2i) - (3 - 4i) equals: (a) 2 - 2i (b) 8 + 6i (c) 2 + 6i (d) 2 - 6i
   Answer: (c)
8. The real part of (2 + i)(2 - i) is: (a) 3 (b) 4 (c) 5 (d) 0
   Answer: (c) - the product is 4 + 1 = 5 + 0i.

Short questions for written practice

1. Define a rational number and give two examples; define an irrational number and give two examples.
2. Verify the associative property of multiplication for a = 2, b = -3, c = 4.
3. Simplify: (a) sqrt(200) (b) $8^{2/3}$ (c) sqrt(27) + sqrt(75)
4. Write 0.00000056 and 78,300,000 in scientific notation.
5. If z = 4 - 3i, find z + (conjugate of z) and z x (conjugate of z).
6. Express (2 + i) / (1 + 2i) in the form a + bi. (Answer: 4/5 - (3/5)i)

> **One-page revision:** R = rational + irrational. Properties: closure, commutative, associative, identity (0 and 1), inverse (-a and 1/a), distributive. Radical: $\sqrt[n]{x} = x^{1/n}$. Scientific notation: $a \times 10^{n}$, $1 \le a < 10$. Complex: a + bi, $i^2 = -1$, conjugate a - bi, divide using the conjugate.

End of Unit 2 notes. Next unit: Unit 3 - Logarithm. Source book: A Textbook of Mathematics for Grade IX, KPTB Peshawar.